

Gao Huxin (高沪昕)

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• WORKING EXPERIENCE

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| 10/2023-now | Postdoctoral fellow at the Chinese University of Hong Kong |
| 02/2023-10/2023 | Research assistant at the Chinese University of Hong Kong |
| 09/2018-12/2018 | Research assistant at National University of Singapore |

• EDUCATION BACKGROUND

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| 01/2019-01/2023 | National University of Singapore
PhD in Biomedical Engineering
CAP: 4.70/5.0 |
| 09/2014-06/2018 | Wuhan University
BEng in Mechanical Engineering
GPA: 3.75/4.0 Ranking: 3/148 |

• RESEARCH INTEREST

Surgical robotics, Robotic simulation, AI applications in surgical robot, Autonomous control

• HONORS & AWARDS

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| 06/2023 | Bronze Award in 9th Hong Kong University Student Innovation & Entrepreneurship Competition |
| 04/2023 | Silver Award in 2023 Inventions Geneva |
| 11/2021 | Silver Award in EMEDIC GLOBAL 2021 (Project: Skullbot, Teamleader) |
| 11/2021 | Best Presentation Award in EMEDIC GLOBAL 2021 (Project: Skullbot, Teamleader) |
| 09/2018 | Excellent Graduate Thesis of Wuhan University |
| 05/2018 | Excellent Graduate of Wuhan University |

• RESEARCH EXPERIENCE

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| 03/2023-now | Translation of Gastrointestinal Endoscopic Surgical Robot <ol style="list-style-type: none">Design industrial version for our GESRImprove mechanical structure |
| 06/2020-05/2022 | Gastrointestinal Endoscopic Surgical Robot (National Key R&D Program of China) <ol style="list-style-type: none">Design a modular flexible manipulator system with variable stiffnessAnalyse robotic kinematics and model the stiffness |
| 12/2020-05/2022 | Surgical Action-driven Visual Attention for Autonomous Endoscope Control <ol style="list-style-type: none">Obtain visual attention point on surgical video combining surgical actionVisual servoing control for daVinci system using surgical action-driven attention |
| 06/2019-12/2021 | AI Application in Brain Intervention Robot <ol style="list-style-type: none">Preoperative motion planning (RCM recommendation) using deep reinforcement learningIntraoperative cable-driven robot control |
| 09/2017-06/2018 | Development of a Portable and Flexible Cable-Driven Exoskeleton Robot <ol style="list-style-type: none">Design mechanical exoskeleton for waist assistanceConduct kinematics analysis and dynamics analysis |

Journal

- [1] **H. Gao**, X. Yang, X. Xiao, X. Zhu, T. Zhang, C. Hou, H. Liu, M. Meng, L. Sun, X. Zuo, Y. Li and H. Ren, "Transendoscopic flexible parallel continuum robotic mechanism for bimanual endoscopic submucosal dissection," **The International Journal of Robotics Research (IJRR, JCR-Q1, IF: 7.5)**, 2024.
- [2] **H. Gao**, X. Yang, T. Zhang, X. Xiao, C. Li, M. Meng, X. Zuo, Y. Li, and H. Ren, "A transendoscopic telerobotic system based on cooperative heterogeneous manipulators for endoscopic submucosal dissection," **IEEE/ASME Transactions on Mechatronics (TMECH, JCR-Q1, IF: 6.1)**, 2024, under review.
- [3] C. Hou, **H. Gao**, X. Yang, G. Xue, X. Zuo, Y. Li, D. Li, B. Lu, H. Ren, L. Sun and H. Liu, "A Piezoresistive-based 3-axial MEMS Tactile Sensor and Its Integrated Surgical Forceps for Gastrointestinal Endoscopic Minimally Invasive Surgery," **Microsystems & Nanoengineering (JCR-Q1, IF: 7.3)**, 2024.
- [4] C. Xu*, Z. Fang*, **H. Gao**, T. Zhang, T. Zhang, P. Liu and H. Ren, "Patient-mounted NeuroOCT for targeted minimally-invasive micro-resolution volumetric imaging in brain in vivo", **Advanced Intelligent Systems (AIS, IF: 6.8)**, 2024.
- [5] S. Su, H. Lan, **H. Gao**, H. Dai and H. Ren, "Enhancing anti-interference of magnetic tracking: A MagRobustNet-based framework with self-supervised anomaly detection and measurements recovery," **IEEE Transactions on Industrial Informatics (TII, JCR-Q1, IF: 11.7)**, 2024.
- [6] X. Yang, **H. Gao**, S. Fu, R. Ji, C. Hou, H. Liu, N. Luan, H. Ren, L. Sun, J. Yang, Z. Zhou, X. Yang, L. Sun, Y. Li, and X. Zuo, "A novel miniature transendoscopic telerobotic system for endoscopic submucosal dissection," **Gastrointestinal Endoscopy (GIE, JCR-Q1, IF: 6.7)**, 2023.
- [7] **H. Gao**, R. Hao, X. Yang, C. Li, Z. Zhang, X. Zuo, Y. Li and H. Ren, "Modeling and compensation of stiffness-dependent hysteresis for stiffness-tunable tendon-sheath mechanism in flexible endoscopic robots," **IEEE Transactions on Industrial Electronics (TIE, JCR-Q1, IF: 7.5)**, 2023.
- [8] L. Qiu, J. Cheng, **H. Gao**, W. Xiong, and H. Ren, "Federated semi-supervised learning for medical image segmentation via pseudo-label denoising," **IEEE Journal of Biomedical and Health Informatics (JBHI, JCR-Q1, IF: 6.7)**, 2023.
- [9] S. Su, S. Yuan, M. Xu, **H. Gao**, X. Yang, and H. Ren, "AMagPoseNet: Real-time 6-DoF magnet pose estimation by dual-domain few-shot learning from prior model," **IEEE Transactions on Industrial Informatics (TII, JCR-Q1, IF: 11.7)**, 2023.
- [10] **H. Gao***, W. Fan*, L. Qiu, X. Yang, Z. Li, X. Zuo, Y. Li, M.Q.Meng and H. Ren, "SAVAnet: Surgical action-driven visual attention network for autonomous endoscope control," **IEEE Transactions on Automation Science & Engineering (T-ASE, JCR-Q1, IF: 5.9)**, 2022.
- [11] X. Yang, **H. Gao**, J. Chen, C. Hou, Z. Zhou, R. Ji, H. Liu, H. Ren, Li. Sun, X. Yang, Y. Li and X. Zuo, "Development and primary evaluation of a digestive endoscopy minimally invasive surgical robot system in endoscopic submucosal dissection: an ex vivo, feasibility study," **Chinese Journal of Digestive Endoscopy**, 2022.
- [12] C. Li, Y. Yan, X. Xiao, X. Gu, **H. Gao**, X. Duan, X. Zuo, Y. Li and H. Ren, "A miniature manipulator with variable stiffness towards minimally invasive transluminal endoscopic surgery," **IEEE Robotics and Automation Letters (RAL, JCR-Q1, IF: 4.6)**, 2021.
- [13] B.S. Yeow, H. Yang, M.S. Kalairaj, **H. Gao**, C.J. Cai, S. Xu, P. Chen and H. Ren, "Deployable serial and parallel structures by untethered magnetic deformations of programmable domain folding and cutting," **Advanced Materials Technologies (JCR-Q1, IF: 6.4)**, 2021.

[14] Z. Yi, **H. Gao**, X. Ji, S.Y. Chong, Y. Mao, B. Luo, C. Shen, S. Han, J.W. Wang, S. Jung, P. Shi, H. Ren and X. Liu, "Mapping drug-induced neuropathy through in-situ motor protein tracking and machine learning," **Journal of the American Chemical Society (JACS, JCR-Q1, IF: 14.4)**, 2021.

[15] L. Zhang, K.S. Kumar, H. Hao, C. J. Cai, H. He, **H. Gao**, S. Yue, C. Li, R.C. Seet, H. Ren and J. Ouyang, "Fully organic compliant dry electrodes self-adhesive to skin for long-term motion-robust epidermal biopotential monitoring," **Nature Communication (NC, JCR-Q1, IF: 14.7)**, 2020.

[16] X. Xiao, **H. Gao**, C. Li, L. Qiu, K. S. Kumar, C. J. Cai, B. S. Bhola, N. K. K. King, and H. Ren, "Portable body-attached positioning mechanism towards robotic needle intervention," **IEEE/ASME Transactions on Mechatronics (TMECH, JCR-Q1, IF: 6.1)**, vol. 25, pp. 1105–1116, April 2020.

Conference

[1] T. Chen*, Q. Lyu*, L. Bai*, E. Guo, **H. Gao**, X. Yang, H. Ren and L. Zhou, "LighTDiff: Surgical endoscopic image low-light enhancement with T-diffusion," in **International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)**, 2024, (Best paper shortlist).

[2] T. Zhang, **H. Gao**, and Hongliang Ren, "Inconstant curvature kinematics of parallel continuum robot without static model," in **2024 IEEE International Conference on Robotics and Automation (ICRA)**, 2024.

[3] **H. Gao***, Z. Zhang*, C. Li, X. Xiao, L. Qiu, X. Yang, R. Hao, X. Zuo, Y. Li and H. Ren, "GESRsim: Gastrointestinal endoscopic surgical robot simulator," in **2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)**, 2022.

[4] **H. Gao**, X. Xiao, L. Qiu, M.Q. Meng, N.K.K. King and H. Ren, "Remote-center-of-motion recommendation toward brain needle intervention using deep reinforcement learning," in **2021 IEEE International Conference on Robotics and Automation (ICRA)**, 2021.

[5] X. Xiao, S. Xu, C. Li, X. Gu, **H. Gao**, M. Meng and H. Ren, "Magnetically-connected modular reconfigurable mini-robotic system with bilateral isokinematic mapping and fast on-site assembly towards minimally invasive procedures," in **2021 IEEE International Conference on Robotics and Automation (ICRA)**, 2021.

Workshop

[1] J. Yu, L. Bai, G. Wang, A. Wang, X. Yang, **H. Gao** and H. Ren. "Adapting SAM for surgical instrument tracking and segmentation in endoscopic submucosal dissection videos." in **2024 IEEE International Conference on Robotics and Automation (ICRA) workshop – C4SR+: Continuum, Compliant, Cooperative, Cognitive Surgical Robotics Systems in the Embodied AI Era**, 2024, (Second place – Best poster award).

[2] **H. Gao**, X. Xiao, X. Yang, T. Zhang, X. Zuo, Y. Li and H. Ren, "A miniature 3-DoF flexible parallel robotic wrist using NiTi wires for gastrointestinal endoscopic surgery," in **2022 IEEE International Conference on Robotics and Automation (ICRA) workshop – Frontiers of Endoluminal Intervention: Clinical opportunities and technical challenges**, 2022.

Patents

[1] A flexible endoscopic surgical robotic system (一种柔性内窥镜手术机器人系统). China, 2023. 202311246382.1

[2] A endoscopic surgical robotic system (一种内窥镜手术机器人系统). China, 2023. 202310595277.2

[3] A measuring device and a measuring method to positioning repeatability and dynamic response for the miniature robotic arm (一种用于控制套管运动的装置及控制套管运动的方法). China, 2021. 202111432441.5

[4] An endoscopic guide wire intervention device based on a dual-channel endoscope (一种基于双通道内镜的内镜下导丝介入装置). China, 2024, 202410165794.0

● RESEARCH SERVICE

Journal: IEEE TII/RAL/TASE/Access, Sensor, Journal of Robotics, Frontiers of Mechanical Engineering, Biomimetic Intelligence and Robotics

Conference: ICRA2021-24, IROS2021-24, BioRob2024, ROBIO, ICRAM

● SKILLS

Robotic Software: Autodesk CAD, SolidWorks, ROS, Coppeliasim (previously V-rep), Gazebo

Programming: Python, Matlab, Lua, C++, C

Machine Learning Architecture: Pytorch, Tensorflow, Matlab AI toolbox, Spinningup, Baseline